TECHNICAL REPORT ON A 400 V SIC ISOLATED HALF BRIDGE GATE DRIVER PCB

1 Introduction

1.1 Overview

This project designs and simulates an isolated half-bridge gate driver using SiC MOSFETS in Kicad 10.0 and LTspice 26.1. The point of an isolated half-bridge gate driver is to switch an output between a positive DC rail and ground in a safe way.

There are several advantages to this design such as galvanic isolation, fast switching and half-bridge control which is useful for motor drives and DC-DC converters.

Texas Instruments UCC21520DW isolated dual-channel gate driver (U1) is used to drive two Wolfspeed C3M0120090D 900 V SiC MOSFETs (Q1 and Q2).

1.2 Technical Background

Two constraints must be imposed to design this:
Galvanic Isolation: Since the circuit operates at 400 V, the low-voltage microcontroller needs to be protected from high-voltage faults and electrical noise caused by switching. The UCC21520DW provides this protection through an internal isolation barrier rated at 5.7 kVrms, which separates the control circuit (GND) from the high-voltage power stage (GNDPWR and SW_NODE). This helps protect the microcontroller and makes the circuit safer and more reliable.

Floating High-Side Potential: The MOSFET needs a positive gate-to-source voltage of about 14 V to 15 V to turn on. However, the low-side and high side MOSFETs are not referenced to the same source. The low-side source is connected to ground while high-side source is connected to the switching node (SW_NODE). Therefore, the high-side gate needs to float above the switching node.

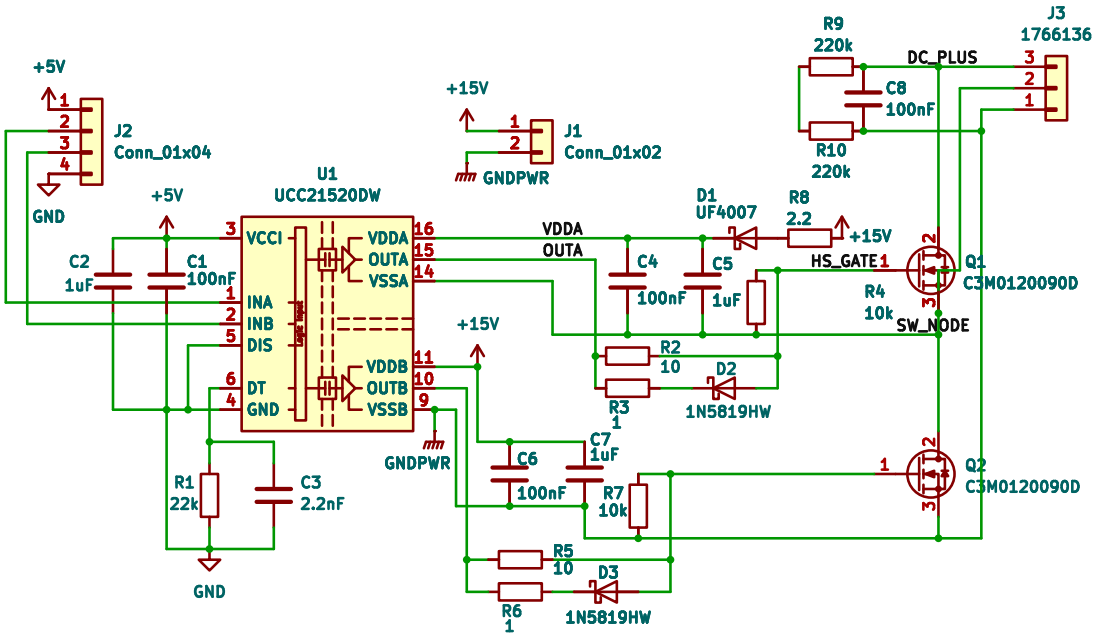


Figure 1 — Schematic of the 400 V SiC half-bridge gate driver made in KiCad; The primary 5 V inputs, the gate driver (U1), the bootstrap charging circuit, split gate resistors and the 400 V power stage is shown.

The table below lists the components used to design the PCB. The components are suited to handling 400 V DC bus safely.

1.3 Bill of Materials and Component Selection

Ref	Qty	Value / Role	Footprint	Manufacturer	Part Number (MPN)	Selection Reasoning
U1	1	Isolated Gate Driver	SOIC-16W	Texas Instruments	UCC21520DW	Wide-body package gives >8 mm creepage between primary and secondary pins;
Q1, Q2	2	900V SiC MOSFET	TO-247-3 Vertical	Wolfspeed	C3M0120090D	The rating is above the safety margin and drain-to-source resistance is low.
D1	1	Bootstrap Diode	DO-41 Axial	Generic	UF4007	Rated for 1000 V with fast 75 ns reverse recovery.
D2, D3	2	Gate Turn-Off Diode	SOD-123 SMD	Generic	1N5819HW	Forward voltage drop is low resulting in a fast turn-off path
C8	1	100nF Snubber Cap	Radial (P=15mm)	WIMA	MKP1J031004B00KSSD	Rated for 630 V and footprint is not too large

R9, R10	2	220kΩ Bleeder	0207 Axial	Yageo	MFR-25FTE52-220K	2 series resistors are used to reduce the voltage across each to 200 V
R1	1	22kΩ Dead-Time	0805 SMD	Yageo	RC0805FR-0722KL	This is to set the hardware dead time on Pin 6 (DT) to approximately 220 ns.
R2, R5	2	10Ω Turn-On	0805 SMD	Yageo	RC0805FR-0710RL	Limits di/dt and control gate charging speed
R3, R6	2	1Ω Turn-Off	0805 SMD	Yageo	RC0805FR-071RL	Turn off path has lower impedance so that Miller bounce is limited
R4, R7	2	10kΩ Gate Pull-Down	0805 SMD	Yageo	RC0805FR-0710KL	MOSFET gates are set to 0V when unpowered, ensuring safety
R8	1	2.2Ω Bootstrap	0805 SMD	Yageo	RC0805FR-072R2L	Controls current while charging
C1, C4, C6	3	100nF Decoupling	0805 SMD	Murata	GRM21BR71H104KA01L	High frequency decoupling near the IC pins
C2, C5, C7	3	1µF Bulk Supply	0805 SMD	Murata	GRM21BR71H105KA12L	The capacitors act like local energy reserves that help to stabilize the IC's supply voltage when it switches fast.
C3	1	2.2nF DT Filter	0805 SMD	Murata	GRM21BR71H222KA01L	Noise filter capacitor is suggested for dead-time resistor by the TI datasheet
J3	1	HV Terminal Header	1x03 (P=7.62mm)	Phoenix Contact	1766246	630 V rated headers for the high voltage terminals
J1, J2	2	Low-Voltage Headers	2.54mm Pitch	Generic	1x02 & 1x04 Male	Standard pin headers to connect to external power sources and microcontroller

2 Design calculations

2.1 Dead time setting

In a half-bridge, if both Q1 and Q2 conduct at the same time, they create a direct short-circuit across the 400 V DC supply, also called a shoot-through, since the drain of Q1 is at the positive DC rail and the source of Q2 is ground. This might burn the MOSFETs. To prevent this, the UCC21520DW includes a programmable dead-time pin (DT, Pin 6) that causes a brief delay between turning one switch OFF and turning the other switch ON.

According to the TI UCC21520DW datasheet, the dead-time (t_DT) is set by the resistor connected from Pin 6 to GND as given below:

$t_{DT} \approx 10\ \text{ns/k}\Omega \times \text{R1}$

Using a standard R1 = 22 kΩ resistor,

$t_{DT} \approx 10\ \text{ns/k}\Omega \times 22\ \text{k}\Omega = 220\ \text{ns}$

A 2.2 nF capacitor (C3) is also placed in parallel with R1 as required by the datasheet to stabilize the pin voltage against switching noise. The delay of 220 ns is wide enough to prevent shoot-through.

2.2 Bootstrap Supply Network (D1, R8, C4, C5)

To keep cost low and circuit simpler, a bootstrap circuit is used instead of a separate power supply. It uses a diode and capacitor to provide the floating voltage. When the low-side MOSFET (Q2) is turned on, SW_NODE goes to 0 V. This allows the +15 V supply to charge the bootstrap capacitors, C4 and C5. The supply goes through the resistor R8 and diode D1, which becomes reverse biased when Q2 turns off and Q1 turns on to block it from discharging back. It is the capacitors that act as a floating battery in this circuit.

High-Side Gate Drive Voltage (V_{boot}): The UF4007 diode (D1) has a voltage drop of $V_F \approx 1.0$ V

Therefore, $V_{boot} = V_{CC} - V_F \approx 15\ \text{V} - 1.0\ \text{V} = 14\ \text{V}$

Voltage Reduction When Turning On Q1 (ΔV_{boot}): From the Wolfspeed C3M0120090D datasheet, the total gate charge required to switch the MOSFET on at 15 V is $Q_g = 17.3$ nC. The capacitor charge equation can be used as follows:

$$\Delta V_{boot} = \frac{Q_g}{C_{boot}} = \left(\frac{17.3 \times 10^{-9}\ \text{C}}{1.1 \times 10^{-6}\ \text{F}}\right) \approx 0.0157\ \text{V (or 15.7 mV)}$$

Since the voltage drop is very small, the 1 μF || 100 nF capacitor combination can hold the supply steady while switching.

Startup Inrush Current Limit (R8): R8 (2.2 Ω) limits the peak charging current from the +15 V rail:

$$I_{peak} = \frac{15\,V - 1.0\,V}{2.2\,\Omega} \approx 6.36\,A$$

As the diode has a peak surge rating of 30 A, the initial current inrush is not an issue.

2.3 Split Gate Resistor Network (R2/R3/D2 and R5/R6/D3)

The gate charging and discharging path are split asymmetrically to control their switching speeds. A Schottky diode is also used so that during turn on, current passes through a larger resistance while during turn off, the diode lets current pass through to discharge faster.

During turn on, the current that charges the gate flows through 10 Ω resistors (R2 and R5). This slows down the rate of current change to prevent excessive voltage spikes and ringing. During turn off, current flows out of the gate through the forward-biased diode (D2 and D3) and the 1 Ω resistor (R3 and R6). In parallel, the equivalent resistance is given by:

$$R_{off} = \frac{1\,\Omega \times 10\,\Omega}{1\,\Omega + 10\Omega} \approx 0.91\,\Omega$$

Resistors R4 and R7 are pull down resistors of 10 kΩ each that connect gate to source. They keep the MOSFETs off when the driver is not powered.

2.4 Decoupling Capacitor and Series Bleeder Resistors (C8, R9, R10)

Decoupling Capacitor (C8 = 100 nF, 630 V) that are between DC plus and ground (GNDPWR) next to the MOSFETs absorb high-frequency transients from switching.

Bleeder resistors allow the high voltage to be discharged after the supply is turned off. Instead of 1 large resistor, two smaller ones are used to divide the voltage as normal through-hole resistors have only a maximum voltage rating of 250V.

$$V_{R9} = V_{R10} = \frac{400\,V}{2} = 200\,V$$

The continuous power dissipated in each resistor at 400 V is:

$$P = \frac{V^2}{R} = \frac{(200\,V)^2}{220k\,\Omega} \approx 0.182 \text{ (below 0.25 W rating)}$$

3 KiCAD PCB Layout

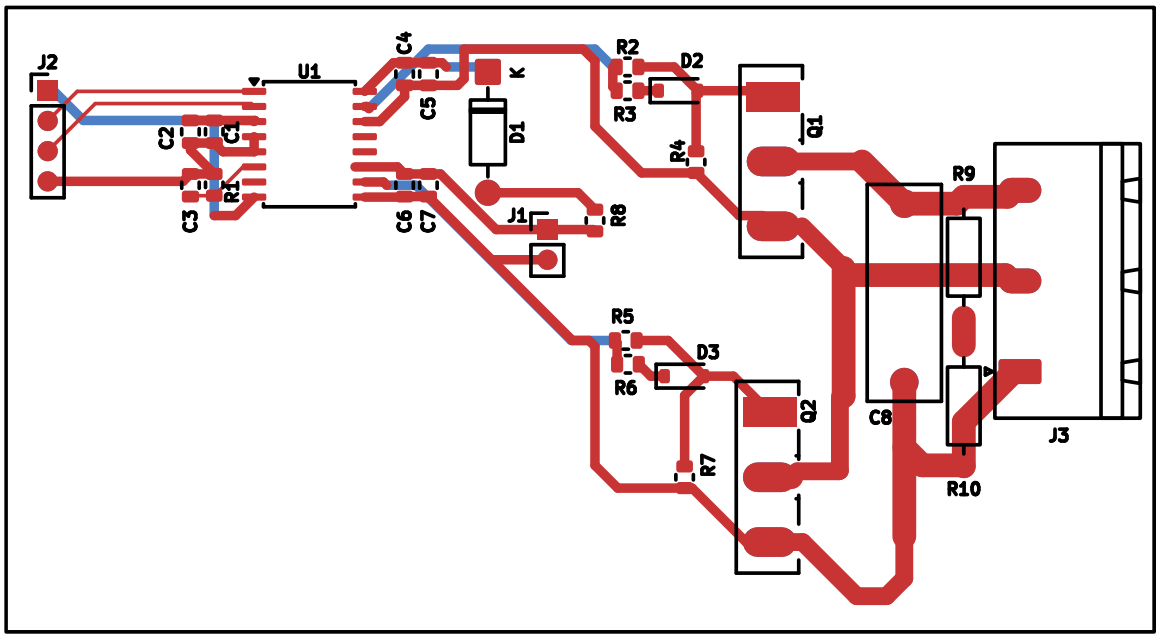


Figure 2 — PCB layout in KiCad. The logic side is on the left while the power stage is on the right. Two layer traces can be seen (red and blue)

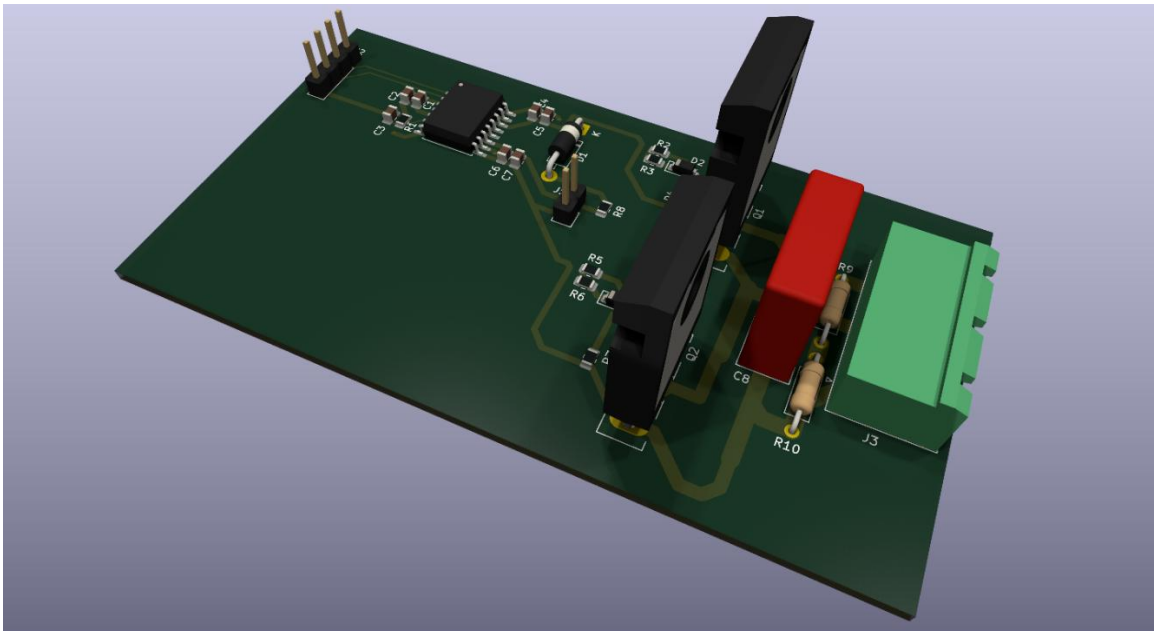


Figure 3 — PCB 3D view: Component spacing is visible

At high voltages such as 400 V, there is a need for sufficient clearance to prevent arcing, either through the board surface or through air. Besides, routing should be compact to reduce stray inductance.

3.1 Separation Between the Logic and High-Voltage Sides

No traces were placed under the UCC21520DW (U1). This helps maintain the required separation between the 5 V logic ground (GND) and the high-voltage side (GNDPWR / SW_NODE). The spacing was kept above 8 mm to improve isolation between the two sides.

3.2 Setting Minimum Clearance in KiCad

To maintain safe spacing around the high-voltage nets, a custom 2.9 mm minimum clearance rule was set up in KiCad. This was applied to DC_PLUS and SW_NODE against other nets. The exception was for connections within the same floating 15 V high-side section, such as VDDA, OUTA, and HS_GATE. The board layout was checked, and the clearance violations were fixed until there were no DRC errors remaining.

3.3 Trace Widths Around the MOSFET Pins

The MOSFET pins are spaced 5.45 mm apart from centre to centre, leaving about 2.6 mm between the edges of adjacent copper pads. If the trace thickness was maintained at 2.0 mm upto the pads, the clearance would be quite low. To avoid this, the DC_PLUS, SW_NODE, and GNDPWR traces were narrowed to 1.5 mm near the MOSFET pads. They were then routed away from the pads before widening back to 2.0 mm. The limited spacing at the pads is due to the package layout.

3.4 Routing SW_NODE and Keeping the Power Stage Compact

To avoid extra vias and reduce high-voltage routing concerns on the bottom layer**, t**he trace was routed entirely on the top layer between C8 Pad 1 and the midpoint of R9 and R10.

C8, R9/R10, and J3 were also placed close to Q1 and Q2 to keep the high-current switching path short and reduce unwanted parasitic inductance.

4 LTspice simulation and verification

4.1 Difference with KiCad Schematic

To verify that the gate drive network, dead-time setting, and 400 V switching stage operate properly, the secondary side of the circuit was simulated in LTspice 26 over a 20 µs transient window (.tran 0 20u 0 1n, covering two full 100 kHz switching periods).

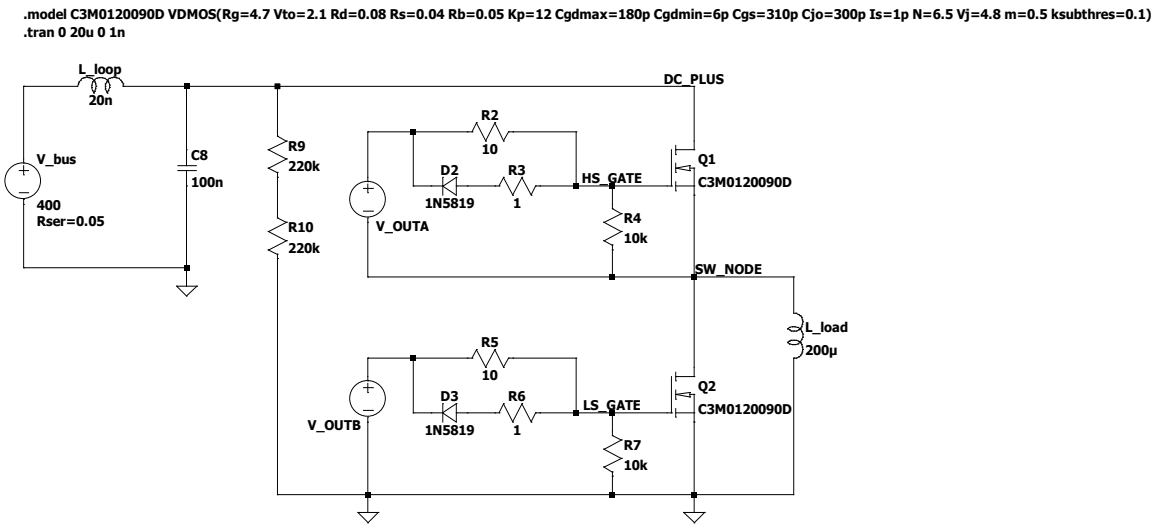


Figure 4 — Modified LTspice schematic for simulating the secondary-side of the gate driver.

In LTspice, some simplifications were made to avoid convergence bugs. The focus of the simulation is to verify that the half-bridge switches between 0 V and 400 V properly, the dead-time prevents shoot-through and the split gate resistors control turn-on and turn-off speed.

4.1.1 UCC21520DW IC and Bootstrap circuit replaced with ideal pulse voltage sources.

Time-step convergences often occur when using vendor ICs. As the internal logic of the IC does not need to be reproduced, the U1 and bootstrap components are replaced by two floating voltage sources that are named V_OUTA and V_OUTB.

The negative terminal of V_OUTA is connected directly to SW_NODE reproducing the effect of connecting to Pin VSSA on the PCB. On the other hand, pulses between 0 V and 14 V (PULSE(0 14 1u 15n 15n 4u 10u)) reproduce the 1 V forward drop lost across bootstrap diode D1 (15 V - 1 V = 14 V).
V_OUTB is referenced to ground (0) and pulses between 15 V and 0 V (PULSE(15 0 0.78u 15n 15n 4.44u 10u)).

Both sources use 15 ns rise and fall times which is realistic and matches the UCC21520DW datasheet. V_OUTB and V_OUTA maintain a 220ns dead-time.

4.1.2 External Bench Supply with Parasitics (V_{bus} and L_{loop}) and Load (L_{load})

To perform the simulation, a 400 V DC supply is connected. Cable inductance is also replicated. Finally, an inductive load is connected to the half-bridge as they model common use of gate drivers.
Parameters:
V_{bus}: 400 V, R_{ser} = 0.05 Ω
L_{loop} = 20 nH, $R_{_ser}$ = 0.05 Ω, R_{par} = 20 Ω
L_{load} = 200 µH, R_{ser} = 20 Ω

$$I_{load,peak} = \frac{V_{bus}}{R_{load}} \times \left(1 - e^{\frac{-t_{on}}{L/R}}\right) = \frac{400\ V}{20\ \Omega} \times \left(1 - e^{\frac{-4\ \mu s}{10\ \mu s}}\right) \approx 6.59\ A$$

4.1.3 Custom SPICE Model for the C3M0120090D MOSFET

A custom SPICE model was created for the Wolfspeed C3M0120090D SiC MOSFET using the VDMOS model in LTspice. The model parameters were selected to represent the

device's main electrical characteristics, including gate resistance, threshold voltage, on-resistance, and gate capacitances. The model also includes parameters for the body diode to represent its forward conduction behaviour.

The main parameters include Vto=2.1 for the gate threshold voltage, Rd=0.08 and Rs=0.04 for the drain and source resistances, and Rg=4.7 for the gate resistance. The capacitance parameters, such as Cgs=310p and Cgdmax=180p, help represent the MOSFET's switching behaviour in the simulation.

4.2 Results

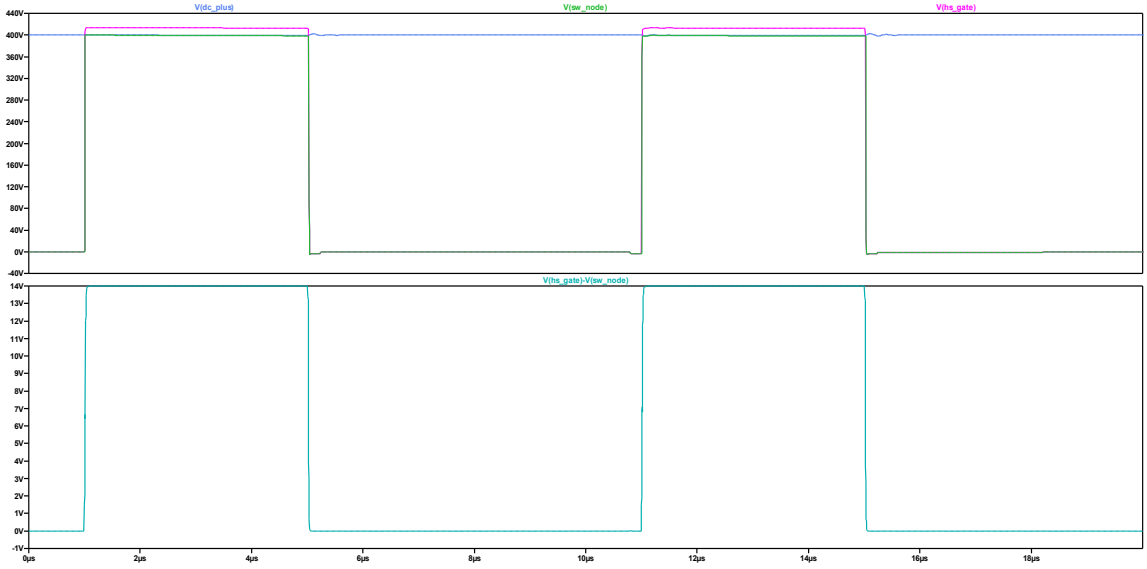


Figure 5 — Waveform result from simulation for two time periods at 100kHz (20 µs). Waveform above shows V(dc_plus) maintaining a stable 400V, V(sw_node) alternating between 0 V and 400 V sharply and the floating voltage V(hs_gate) reaching 414 V. Waveform below shows the difference between V(hs_gate) and V(sw_node) switching between 0 V to 14 V.

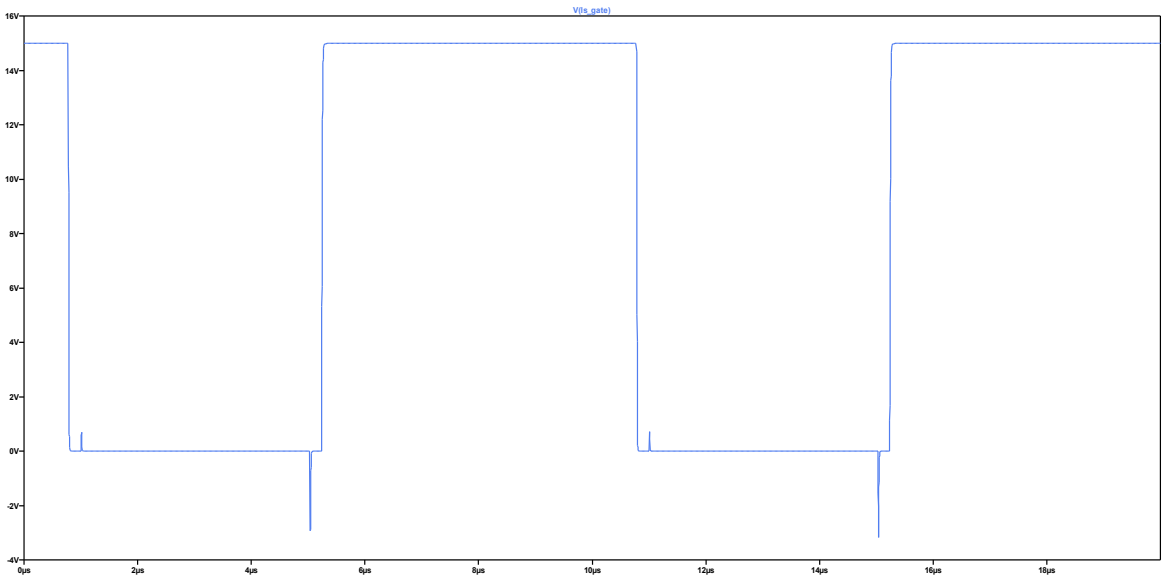


Figure 6 — The low-side gate-to-source voltage waveform shows the MOSFET switching at 15 V, with a 220 ns dead time between switching events. Small voltage spikes appear at 1 µs and 11 µs due to the Miller effect (below the 1.8 V turn-on threshold).

4.2.1 Findings and Discussion

1. The switching node [V(sw_node)] can switch properly from 0 V to 400 V and no shoot-through appears.
2. DC Bus is maintained at a stable 400 V with minimal ringing. Capacitor C8 absorbs the switching edge.
3. High-side floating gate is 14V above SW_NODE when OUTA is HIGH, going up to 414 V when SW_NODE is at 400V.
4. On V(ls_gate), the rising edge is smoother than the falling edge. This is because of the 10 Ω resistor R5 which slows down current while rising while the R6 and D3 are what makes the falling edge sharp. When Q1 switches on at 1.0 µs and SW_NODE reaches 400V, the Miller bounce on V(ls_gate) peaks at 0.7-0.8 V.

5 Design Limitations and Future Improvements

After reviewing the board and analysing the simulation, three compromises were identified that could be addressed in a newer version of the board

(a) Shorter Trace Path and Compact Component Placing:

In this version of the board, components are placed with more than sufficient spacing to make routing easier and to inspect visually which made some traces, particularly between the gate driver (U1), MOSFETs (Q1 and Q2) and the decoupling capacitor (C8) longer. This introduces more stray inductance that can increase voltage ringing during switching. This can be improved by moving components closer, overlapping 2-layer traces and replacing some tracks with copper polygon pours.

(b) Bipolar Gate Drivers and Full 15 V High-Side Supply:

The SiC MOSFET has low minimum gate threshold voltage of 1.8 V. While the simulated gate bounce was around 0.8 V, in practice, higher voltage bounce may occur causing unintentional turn-on. Furthermore, the 1.0 V forward voltage drop across the diode D1 lowers the high-side gate drive voltage to ~14 V. Replacing the bootstrap circuit with small isolated bipolar DC-DC power supply (-4 V / +15 V) would solve these issues.

(c) Reducing Dead-Time Conduction Losses:

In this version, the dead-time is 220 ns which is quite generous to ensure zero shoot-through but wastes extra power as heat as a result. Therefore, changing R1 from 22 kΩ to 10 kΩ will shorten the dead-time to about 100 ns, improving efficiency.